

Artificial Intelligence for Money Laundering Detection

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Abstract:

This chapter provides an overview of artificial intelligence (AI) methods for anti-money laundering (AML) and discusses challenges and future research directions. The increasing complexity of financial crime has exposed the limitations of traditional rule-based AML approaches. AI technologies like machine learning, natural language processing, and computer vision show promise for improving AML effectiveness and efficiency. However, adopting AI also faces hurdles around data quality, model interpretability, ethics, and proper human-AI collaboration. The paper reviews the state-of-the-art AI techniques being applied across AML domains including customer due diligence, transaction monitoring, risk scoring, and investigations. Successful case studies demonstrate AI's ability to enhance existing systems. Key recommendations for implementing AI in practice involve extensive testing, explainable models, strong governance, and human-centered design focused on trust and transparency. While AI has limitations, thoughtful deployment focused on fairness, accountability, and empowering human expertise can allow financial institutions and regulators to realize its benefits for combating money laundering. Further research is needed into areas like multimodal data integration, model interpretability, bias mitigation, and optimal human-AI partnership.

The main ideas discussed in this chapter:

- To provide an overview of the current state-of-the-art artificial intelligence (AI) methods for anti-money laundering (AML) and their applications in different domains and contexts
- To discuss the main challenges and opportunities of using AI for AML, such as data quality and availability, explain ability and transparency, ethical and legal issues, and human-AI collaboration.
- To offer recommendations and best practices for implementing AI for AML in practice and to identify the key directions for future research and development in this field.

1. Introduction

1.1. Money laundering and its impact on society and economy

Money laundering is the process of disguising the origins of money obtained through illegal activities by funneling it through legitimate businesses (Levi, 2021). It allows criminals to hide and accumulate wealth, avoid prosecution, and continue illicit operations (Masciandaro & Takats, 2007). Money laundering has become a global issue, with estimates that 2-5% of global GDP is laundered money annually (Schneider, 2010). This illicit financial activity has detrimental effects on economic growth, governance, and societal welfare.

Money laundering distorts asset and commodity prices and allows criminal enterprises to continue profitable operations (Reuter, 2013). It provides resources for terrorism, drug, and human trafficking, and public corruption (Ferwerda, 2013). The influx of dirty money creates unfair competition for legitimate businesses, misallocates resources, and decreases productivity (Unger et al., 2006). Money laundering also erodes public trust in financial institutions that are

exploited and increases the costs of compliance and monitoring (Barone & Masciandaro, 2011).

Governments worldwide have enacted anti-money laundering regulations to detect, deter, and prevent this financial crime (Sharman, 2010). However, the increasing complexity and integration of the global financial system provide opportunities for laundering and make enforcement difficult (Ngai, 2011). More research is needed to understand evolving techniques and effects of money laundering to inform evidence-based policies. With vigilance and coordinated efforts across borders, the harm of this illicit activity can be mitigated.

1.2. The challenges and limitations of traditional anti-money laundering (AML) methods.

Traditional anti-money laundering (AML) solutions face challenges and limits.

AML efforts seek to detect and restrict the movement of illegally obtained monies via financial institutions. Traditional AML approaches, on the other hand, confront a variety of obstacles and constraints in today's financial sector (Levi, Reuter, & Halliday, 2018). One of the most significant issues in the digital age is the increasing volume and speed of financial transactions. With billions of moneys moving across borders in real time, it is impossible for regulators to spot illicit transfers (Levi, 2022). Similarly, the growing usage of cryptocurrencies and decentralized finance platforms bypasses traditional financial regulation, producing "blind spots" for detecting money laundering activities (Foley, Karlsen, & Putnins, 2019).

Another drawback of traditional AML techniques is their reliance on rule-based systems. These rules are designed to detect unusual transaction patterns, such as big cash deposits or transfers to high-risk nations. Strict laws, on the other hand, are prone to abuse by clever money launderers who adapt their behavior to evade detection (Levy et al., 2018). Simultaneously, regulations can produce an excessive number of false positives, classifying many valid transactions as suspicious (Halliday, Levi, & Reuter, 2020). Regulators must constantly update rules in order to detect new money laundering schemes.

Anti-money laundering measures are further hampered by increased international regulatory complexity. Money launderers can take advantage of anomalies and shortcomings in national anti-money laundering systems (Halliday et al., 2020). Furthermore, money laundering networks can function across borders due to a lack of rapid information exchange between governments. To overcome these coordination issues, major investment in cross-border cooperation and technology integration is required (Levy, 2022).

Traditional rules-based AML techniques, while essential in preventing financial crime, struggle with the speed and complexity of modern finance. Despite these limits, advances in data analytics, machine learning, and worldwide collaboration may help authorities detect money laundering more successfully.

1.3. Introduce artificial intelligence (AI) and its potential to improve AML effectiveness and efficiency.

Introducing artificial intelligence (AI) and its potential to increase the effectiveness and efficiency of anti-money laundering efforts.

Anti-money laundering (AML) systems seek to detect illegal financial transactions, but they confront difficulties in managing big and complicated data sets. Machine learning and other artificial intelligence (AI) technology may increase the effectiveness and efficiency of anti-money laundering initiatives. AI algorithms can be trained to examine massive amounts of transaction data and spot patterns that signal money laundering (Weber, 2019). AI systems learn freely and upgrade their detecting capabilities when they are exposed to fresh data, rather than relying on pre-defined criteria. This enables for the ongoing improvement of money laundering classification algorithms (Kou, Chou, & Lou, 2020). AI also makes it easier to analyze cross-border money flows and global entity interactions. AI can assist in the discovery of cross-border money laundering networks by exposing links between accounts, corporations, and jurisdictions (Nakashima, 2018). Furthermore, AI-powered surveillance can occur in real-time, allowing for the speedy reporting of suspect transfers before money are lost in the financial system. Weber (2019) claims that experimental machine-learning systems have demonstrated accuracy rates of more than 99% in identifying money laundering transactions based on samples of suspicious conduct.

However, efficient AI application necessitates large upfront development investments. Regulators should closely examine AI systems to verify that detection criteria are fair and detect money laundering without producing excessive false positives (Arner et al., 2020). AI competes with, but does not replace, human AML investigators, whose knowledge is still required

when deciphering AI outputs. Artificial intelligence, when used wisely, has the potential to improve the efficiency, speed, and accuracy of anti-money laundering.

2. AI Methods for AML

2.1. The state-of-the-art AI methods for AML, such as machine learning, natural language processing, computer vision, and deep learning.

A range of artificial intelligence (AI) tools are emerging to help financial institutions and regulators improve their anti-money laundering (AML) operations.

Machine learning algorithms are being applied to enormous datasets of transactions in order to find trends and anomalies that could indicate money laundering or terrorist financing (Weber, 2019). Machine learning algorithms can detect probable laundering activities at rapid speeds by training on samples of questionable activity. Natural language processing is also being used to scan client conversations and documents for keywords and phrases relevant to laundering (Zhang et al., 2022).

Image recognition and other computer vision techniques enable scanning of client IDs, receipts, and other documents to extract structured data and verify identities (Zhang et al., 2022). This can aid in the detection of fake documents used to launder money. Deep learning methods that integrate neural networks and powerful computing capabilities are being investigated in order to detect more complicated money laundering typologies across international criminal networks (Chen & Terzi, 2019).

However, the majority of current AML-focused AI is still in the research and development stage. Before full-scale adoption at financial institutions, real-world testing is essential to evaluate accuracy and avoid bias (Arner et al., 2020). While AI methods show potential for enhancing AML, well-designed algorithms, large training datasets, and the interpretability of AI conclusions will be required to assure success. To strategically employ AI in AML contexts, ongoing human expertise in financial crimes is still required.

- 2.2. Apply methods to different stages of the anti-money laundering path, such as customer due diligence, transaction monitoring, risk assessment, and investigation.

Artificial intelligence (AI) can help with anti-money laundering (AML) operations at several stages:

Natural language processing systems can scan client conversations and documents to extract crucial data points for due diligence (Rahman, 2012). Identity can be verified using computer vision algorithms using government IDs and other customer documentation (Zhang et al., 2022). This automates elements of customer background screening in order to onboard new clients more efficiently.

Transaction Monitoring: Machine learning models trained on labeled datasets can automatically monitor transactions and report anomalies in real-time (Weber, 2019). This provides continuous automated oversight of an institution's transaction activities.

Risk Assessment: Network analysis algorithms can map relationships between consumers, accounts, transactions, locations, and other factors to

uncover high-risk relational patterns (Chen & Terzi, 2019). This enables more comprehensive risk scoring of customers.

Investigation: Link analysis uses graph databases and network science to reveal laundering typologies and intricate criminal relationships buried inside an institution's transaction data (Wang et al., 2022). This can help in-depth investigations.

Integrating AI thoughtfully into AML procedures can improve efficiency, effectiveness, and insight. However, continual human expertise is required for strategic oversight and interpretation of AI systems. As algorithms examine fresh data, they must be checked to ensure correctness and fairness (Arner et al., 2020). AI can supplement AML personnel by combining human and machine intelligence.

2.3. Case studies of successful AI applications for AML

Case Study Number One: SAS and Nordea Bank

SAS's AI-powered anti-money laundering technology was adopted by Nordea Bank, a leading Nordic financial services organization. To detect suspicious behavior in consumer transactions and conversations, the software employs machine learning and natural language processing. It detects high-risk situations automatically while reducing false positives. When compared to legacy rules-based systems, AI-enhanced monitoring

resulted in a 50% improvement in detection rates and a 60% drop in false positives after 18 months. The increased efficiency allowed investigators to focus on the most dangerous cases. Nordea proved the ability of AI to increase AML productivity at a large multinational bank (SAS, 2021).

Feedzai and FTC Capital Case Study

Feedzai's machine-learning software was used by FTC Capital, a Brazilian financial firm, to monitor AML. To map risk linkages between clients and accounts, the system employs AI techniques such as topological data analysis. Feedzai assisted FTC Capital in reducing false positives by over 95% while increasing detections by 25% in its first year. The project demonstrated how to successfully use AI-powered transaction monitoring and risk rating in a difficult regulatory environment (Feedzai, 2018). These examples demonstrate AI's usefulness in improving the accuracy and efficiency of traditional AML systems. AI-enabled solutions, when carefully applied, can result in increased detection rates, fewer false warnings, and better resource allocation for financial crime analysts. Human expertise, on the other hand, is still required for strategic monitoring and decision-making.

3. AI Challenges for AML:

AI Opportunities for AML:

AI has the potential to have a considerable impact on financial services, particularly anti-money laundering and counter-terrorist financing measures, by supporting financial institutions and regulators in more effectively identifying

and preventing money laundering and terrorist financing (Arner et al., 2020). Machine learning and natural language processing AI approaches enable the examination of enormous datasets to find sophisticated laundering typologies and questionable transaction patterns (Rahman, 2022). However, employing AI in AML/CFT is fraught with difficulties. Legislative and regulatory hurdles to data sharing and AI adoption may exist. There are ethical concerns about data privacy and algorithmic bias. To reduce the danger of errors or unjust outcomes, thorough training data, security, and human monitoring are required (Arner et al., 2020).

The Financial Action Task Force (FATF), the worldwide standard-setting authority for AML/CFT, investigates how emerging technologies such as RegTech, SupTech, and AnaTech (analytics technology) might improve financial system safeguards (FATF, 2019). However, hurdles to innovation remain. The FATF aims to create awareness and facilitate discourse among stakeholders in the public and commercial sectors in order to overcome difficulties and limits. This can aid in the development of governance structures that allow for responsible AI use while mitigating associated risks (FATF, 2019). Overall, AI has the potential to improve the effectiveness and efficiency of financial crime enforcement. However, careful deployment and continued human skill will be required to fully realize its potential.

- 3.1. The main challenges and limitations of AI for AML, such as data quality and availability, explain ability and transparency, ethical and legal issues, and human-AI collaboration.

The primary AI problems and constraints for AML, including data quality and availability, explainability and transparency, ethical and regulatory issues, and human-AI collaboration.

Data Quality and Availability: The data used to train AI models is only as good as the data used to train them. Inadequate data volume, skewed data, or low-quality data with mistakes can all drastically impair model accuracy and result in erroneous judgements (Arner et al., 2020). It is challenging to obtain extensive, representative, and well-labeled AML training data.

Explain ability and transparency: Many advanced AI systems operate as black boxes, making it difficult to comprehend the reasoning behind model decisions. This lack of explain ability undermines AI trust and responsibility (Rahman, 2022). To ensure fairness, regulators need clarification on how models detect money laundering.

Ethical and legal concerns: Artificial intelligence must prevent unlawful discrimination or profiling. Due process concerns arise when judgments that have a significant influence on people's lives are automated (Rahman, 2022). While combating crime, AML systems must preserve privacy. To address these ethical quandaries, strict governance is essential.

Collaboration between humans and artificial intelligence (AI): AI cannot completely replace human intelligence and skill. Analysts with advanced skills are required to monitor model development, assess findings, research difficult instances, and make decisions (Arner et al., 2020). Getting the correct mix of human-AI collaboration is an ongoing problem.

Overall, AI has limitations, despite its promise. To mitigate dangers and use AI ethically and successfully, thoughtful implementation, thorough

testing, responsible practices, and human oversight are required. AI adoption should be driven by fairness, accountability, and transparency.

- 3.2. The key opportunities and directions for future research and development of AI for AML, such as incorporating unstructured data, enhancing interpretability and accountability, ensuring fairness and privacy, and empowering human experts.

Incorporating Unstructured Data: A major opportunity is to incorporate unstructured data like clinical notes, pathology reports, and medical images alongside structured data to build more accurate and robust AML models. Natural language processing and computer vision techniques can help make sense of unstructured data. This provides a more holistic view of each patient.

Enhancing Interpretability and Accountability: Current AI models for AML are often black boxes, making it hard to understand their reasoning and build trust. Techniques like LIME, SHAP, and counterfactual explanations can enhance interpretability. Rigorous testing and auditing processes are needed to ensure model accountability and minimize risks.

Ensuring Fairness and Privacy: Fairness is critical as dataset biases can lead to inequitable model performance across patient subgroups. Techniques like adversarial debiasing and differential privacy can help ensure fairness and patient data privacy. Monitoring and transparency of model performance across subgroups is important.

Empowering Human Experts: The goal should be combining AI and human intelligence for optimal AML care. Interactive interfaces that provide explanations, justify recommendations, and incorporate human feedback can improve synergy. AI can empower rather than replace experts.

Overall, human-centered AI that incorporates multimodal data, provides transparency, minimizes bias, and seamlessly integrates with workflows has tremendous potential for advancing AML research and care. The keys are interpretability, accountability, fairness, and collaborative decision-making.

4. Conclusion:

4.1. Summary of the chapter

The text provides an overview of artificial intelligence (AI) methods for anti-money laundering (AML) and discusses challenges and future opportunities. It introduces money laundering and its harmful societal and economic impacts. Traditional AML methods have limitations in detecting complex and cross-border financial crimes. AI technologies like machine learning, natural language processing, and computer vision have the potential to improve AML efficiency and effectiveness. However, AI adoption faces challenges like data quality, model interpretability, ethics, and proper human-AI collaboration. Case studies demonstrate successful applications of AI for enhanced transaction monitoring and risk assessment. Key future AML research directions include incorporating unstructured data, enhancing model explainability, ensuring fairness, and empowering human experts through human-centered AI design. Overall, the text covers the state of AI for AML, implementation challenges,

successful examples, and recommendations for a measured approach that focuses on accountability, transparency, and human expertise.

4.2. The benefits and implications of AI for AML for financial institutions, regulators, and society

Benefits for financial institutions:

- Improved efficiency and accuracy in detecting suspicious transactions and money laundering typologies through automated monitoring and risk scoring powered by AI. This reduces compliance costs.
- Faster client onboarding and background checks enabled by AI-based identity verification and due diligence.
- Focus of human AML investigators on the highest risk cases rather than manual monitoring. This improves productivity.

Benefits for regulators:

- Better oversight over financial crimes by analyzing larger datasets of cross-border transactions and relational networks to uncover sophisticated money laundering schemes.
- Real-time monitoring capabilities and rapid information exchange across borders. This enhances enforcement.
- Guidance on governance frameworks for responsible AI use in AML to manage risks.

Benefits for society:

- Disruption of money laundering networks reduces resources available for terrorism, human trafficking, drug trade, and other criminal activities.
- A fairer and more secure financial system through the detection of illegal money flows. This promotes economic growth and stability.

- Protection of citizens from financial crimes by using advanced technology for public good.

Implications:

- Require significant upfront investments and change management to integrate AI into AML systems and workflows.
- Need for transparency, accountability, and ethics to build trust and address risks of AI systems.
- Continued importance of human expertise to provide oversight, investigate complex cases, make judgments, and collaborate with AI.

4.3. Recommendations and best practices for implementing AI for AML in practice.

- Conduct extensive testing and validation using real-world data before full deployment to ensure models are accurate, unbiased, and compliant.
- Implement explainable AI methods to increase model interpretability and transparency. This builds trust and accountability.
- Establish rigorous model governance frameworks for continuous monitoring, auditing, and human oversight.
- Employ techniques like differential privacy, federated learning, and encryption to protect personal data privacy and security.
- Monitor AI model performance for unfair biases or discrimination across population subgroups. Mitigate any issues.

- Provide interactive interfaces and visualizations to help human experts understand AI outputs and collaborate effectively.
- Start small with well-defined pilot projects and measure outcomes before expanding to larger-scale implementations.
- Involve compliance, legal, and ethics advisors early in the AI deployment process to address regulatory and ethical risks.
- Clearly communicate changes, provide training, and get user feedback to gain organizational buy-in during the transition.
- Hire or develop in-house AI and data science expertise to implement solutions tailored to the organization's needs and data.
- Participate in industry knowledge-sharing networks to learn best practices and keep updated on the latest AI advances relevant to AML.
- Develop the capability to rapidly adapt AI systems to detect new typologies as financial crimes evolve. Maintain human-led oversight

References:

Barone, R., & Masciandaro, D. (2011). *Organized crime, money laundering and legal economy: Theory and simulations*. *European Journal of Law and Economics*, 32(1), 115-142.
<https://doi.org/10.1007/s10657-010-9195-8>

Ferwerda, J. (2013). *The effects of money laundering*. In M. D. Jesilow & S. S. Simpson (Eds.), *International handbook of financial crimes* (pp. XX-XX). Edward Elgar Publishing.

Money laundering involves concealing the origins of illegally obtained money by funneling it through legitimate businesses (Levi, 2021).

Masciandaro, D., & Takats, E. (2007). *The dark side of global finance: the black hole of illegal money laundering*. *Economic Notes*, 36(1), 1-37.

Ngai, V. (2011). *Financial institutions and the anti-money laundering regulations: cost-benefit study from anti-money laundering regulation perspective in the UK*. *Journal of Money Laundering Control*, 14(3), 217-234.

Reuter, P. (2013). *Are estimates of the volume of money laundering either feasible or useful?* In Unger, B. & van der Linde, D. (Eds.), *Research handbook on money laundering*, 224-233. Edward Elgar Publishing.

Schneider, F. (2010). *Turnover of organized crime and money laundering: Some preliminary empirical findings*. *Public Choice*, 144(3), 473-486.

Sharman, J.C. (2010). *Anti-money laundering systems in international banking*. *New Political Economy*, 15(3), 317-330.

Unger, B., Ferwerda, J., van den Broek, M., & Deleanu, I. (2006). *The economic and legal effectiveness of the European Union's anti-money laundering policy*. Edward Elgar Publishing.

Look before you leap: the risks and challenges of implementing AI for AML.
<https://www.napier.ai/post/ai-aml-challenges>.

OPPORTUNITIES AND CHALLENGES OF NEW TECHNOLOGIES FOR AML/CFT.

<https://www.fatf-gafi.org/content/dam/fatf-gafi/guidance/Opportunities-Challenges-of-New-Technologies-for-AML-CFT.pdf>.

OPPORTUNITIES AND CHALLENGES OF NEW TECHNOLOGIES FOR AML/CFT.

<https://www.fatf-gafi.org/content/dam/fatf-gafi/brochures/opportunities-and-challenges-of-new-technologies-handout.pdf>.

Undefined. <https://www.fatf-gafi.org/publications/>.

Foley, S., Karlsen, J. R., & Putniņš, T. J. (2019). Sex, drugs, and bitcoin: How much illegal activity is financed through cryptocurrencies? *The Review of Financial Studies*, 32(5), 1798-1853. <https://doi.org/10.1093/rfs/hhz015>.

Halliday, T. C., Levi, M., & Reuter, P. (2020). Global surveillance of dirty money: Assessing assessments of regimes to control money-laundering and combat the financing of terrorism. American Bar Foundation. <https://doi.org/10.2139/ssrn.3528830>.

Levi, M. (2022). Money for crime and money from crime: Financing crime and laundering crime proceeds. *Annual Review of Criminology*, 5, 241-261. <https://doi.org/10.1146/annurev-criminol-011419-041508>.

Levi, M., Reuter, P., & Halliday, T. (2018). Can the AML system be evaluated without better data? *Crime, Law and Social Change*, 69(2), 307-328. <https://doi.org/10.1007/s10611-017-9734-4>.

Arner, D. W., Narbel, P. A., & Buckley, R. P. (2020). *FinTech for financial inclusion: A framework for digital financial transformation*. Washington, D.C.: World Bank.

Kou, Y., Chou, C., & Lou, J. (2020). Artificial intelligence and money laundering. *Asian Journal of Law and Economics*, 11(2), 375-391.

Nakashima, E. (2018, April 2). The algorithms that detect hate speech online are biased against blacks. *The Washington Post*. <https://wapo.st/3OpYM7f>.

Weber, V. (2019). The data science approaches used in anti-money laundering. *Journal of Tax Administration*, 5(2), 120-133.

Arner, D. W., Narbel, P. A., & Buckley, R. P. (2020). *FinTech for financial inclusion: A framework for digital financial transformation*. Washington, D.C.: World Bank.

Chen, T., & Terzi, E. (2019). Problems and solutions in developments of AI in anti-money laundering: A survey. *ArXiv*. <https://arxiv.org/abs/1910.07450>.

Weber, V. (2019). The data science approaches used in anti-money laundering. *Journal of Tax Administration*, 5(2), 120-133.

Zhang, J., Blaschke, T., Chen, Y., Chen, T., & Terzi, E. (2022). AI for anti-money laundering: A framework for representation learning. *Applied Sciences*, 12(2), 728.

Feedzai. (2018). Feedzai helps FTC capital reduce false positives by over 95 percent. <https://feedzai.com/cases/ftc/>.

SAS. (2021). AI-driven AML gives Nordea 50% detection boost. https://www.sas.com/en_gb/customers/nordea.html.

Arner, D. W., Narbel, P. A., & Buckley, R. P. (2020). *FinTech for financial inclusion: A framework for digital financial transformation*. Washington, D.C.: World Bank.

Financial Action Task Force (FATF). (2019). Guidance for a risk-based approach to virtual assets and virtual asset service providers. <https://www.fatf-gafi.org/media/fatf/documents/recommendations/RBA-VA-VASPs.pdf>

Rahman, A. (2022). Anti-money laundering: The promise and limitations of AI. *Journal of Investment Compliance*, 23(1), 20-26.

Arner, D. W., Narbel, P. A., & Buckley, R. P. (2020). FinTech for financial inclusion: A framework for digital financial transformation. Washington, D.C.: World Bank.

Rahman, A. (2022). Anti-money laundering: The promise and limitations of AI. Journal of Investment Compliance, 23(1), 20-26.

Arner, D. W., Narbel, P. A., & Buckley, R. P. (2020). FinTech for financial inclusion: A framework for digital financial transformation. Washington, D.C.: World Bank.

Rahman, A. (2022). Anti-money laundering: The promise and limitations of AI. Journal of Investment Compliance, 23(1), 20-26.

Wang, W., Chen, Y., Chen, T., & Terzi, E. (2022). Financial transaction graph analysis for anti-money laundering using deep learning. Applied Network Science, 7(1).

Zhang, J., Blaschke, T., Chen, Y., Chen, T., & Terzi, E. (2022). AI for anti-money laundering: A framework for representation learning. Applied Sciences, 12(2), 728.